

EXPLAINABLE AI-INNOVATE X



-By

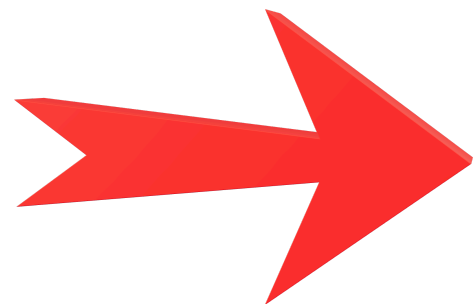
Harish.K - 3rd Year (25508866)

Hariprakash.P - 3rd Year (25508865)

Diploma in Computer Engineering



EXPLAINABLE AI



THE FOUNDATION

What is Artificial Intelligence

AI enables computers to perform tasks that normally require human intelligence.

- Learns patterns from data
- Makes predictions, classifications, and decisions
- Powers chatbots, face recognition, voice assistants, and recommendation systems



DEFINITION

What is Explainable AI (XAI)?



- 1 XAI stands for Explainable Artificial Intelligence.
- 2 It makes AI decisions easier for humans to understand.
- 3 XAI answers the question: “Why did the AI make this decision?”.
- 4 It shows the important factors behind a prediction.



MOTIVATION

Why is XAI **IMPORTANT** ?



Trust

People can trust AI more when they understand its decisions.



Debugging

Developers can identify errors in AI models.



Fairness

XAI can help identify unwanted bias in decisions.

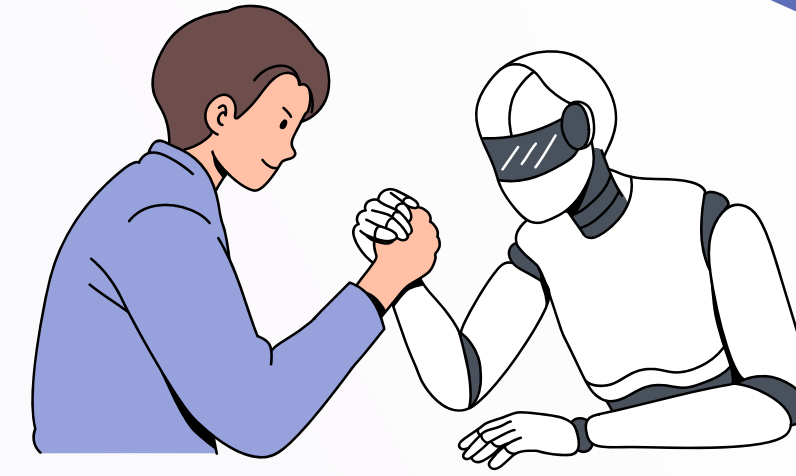


Safety

Understanding AI decisions is important in critical applications.

COMPARISON

Black Box AI vs Explainable AI



BLACK BOX AI

Input → AI Model → Result

- Gives a result.
- Reason behind the result may be difficult to understand.

Example:

“Loan rejected.”

Question: “Why was it rejected?”

EXPLAINABLE AI

Input → AI Model → Result + Explanation

- Gives the result.
- Also provides important reasons.

Example:

“Loan rejected because of low income and repayment history.”

PROCESS

How Does XAI Work?

- 1 Collect Data
- 2 Train AI Model
- 3 Make Prediction
- 4 Generate Explanation
- 5 Human Understands Decision



Common XAI Techniques



Feature Importance

Shows which input features had the biggest influence on the prediction.



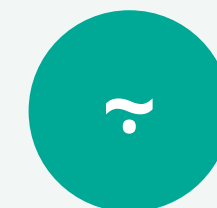
LIME

Explains an individual prediction using a simpler model.



SHAP

Shows how much each feature contributes to the final prediction.



Decision Trees

Uses a tree-like structure to show a sequence of decisions.



WORKED EXAMPLE

Student Marks Prediction

Suppose an AI predicts a student's mark as 82/100.

INPUT FEATURES

- Study hours
- Attendance
- Previous marks
- Assignment score

AI PREDICTION

82

out of 100

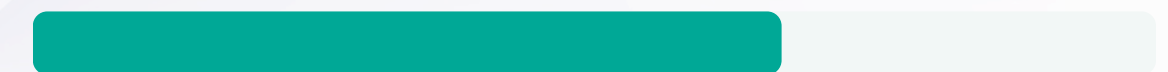
XAI EXPLANATION

Study hours



High impact

Attendance



Positive impact

Previous marks



Medium impact

XAI explains why the AI predicted 82 marks.



REAL-WORLD USE

Applications of XAI



Healthcare

Helps doctors understand AI-supported predictions.



Banking

Can explain credit and fraud-related decisions.



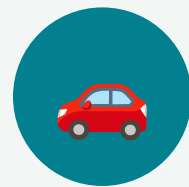
Education

Can explain student performance predictions.



Cybersecurity

Helps analysts understand why activity is flagged.



Autonomous Systems

Can provide reasoning around system decisions.



Business

Makes AI predictions and recommendations easier to evaluate.

Advantages and Limitations

ADVANTAGES

- ✓ Builds trust in AI
- ✓ Helps detect model errors
- ✓ Supports fairness
- ✓ Makes AI easier to audit
- ✓ Helps users make informed decisions

LIMITATIONS

- ! Some explanations are only approximations
- ! Complex AI models can be difficult to explain
- ! Different XAI methods may produce different explanations
- ! An explanation does not automatically mean the AI is correct

CASE STUDY

Real-Life Example



✗ WITHOUT XAI

AI System:
“Transaction flagged as suspicious.”

User: “Why?”

Problem:

The reason may not be clear.

✓ WITH XAI

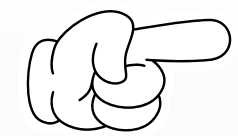
AI System: “Transaction flagged as suspicious.”

Explanation:

- Unusual transaction amount
- New location
- Unusual transaction time

Result: The analyst can understand and review the decision more easily.

CONCLUSION



Key Points

- ✓ XAI means Explainable Artificial Intelligence
- ✓ It explains why an AI system made a particular decision
- ✓ XAI improves trust, transparency, fairness, and debugging
- ✓ It is useful in healthcare, banking, education, cybersecurity, and many other fields





QUESTIONS & ANSWERS



THANK YOU

